



ITSimplera Solutions

WEEK 4 INTERNSHIP TASK REPORT

Unsupervised Learning & Customer Segmentation

K-Means & Hierarchical Clustering on Credit Card Behavioural Data

8,950 Customers Analyzed	18 → 17 Features (post cleanup)	k = 4 Optimal Clusters	2 Algorithms Compared
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1. Overview

The report documents Week 4 of the AI/ML internship at ITSimplera Solutions, focused on unsupervised learning. Unlike the supervised tasks completed in prior weeks, this task has no target column to predict the objective is to discover natural, hidden groupings within customer behaviour and translate them into segments a bank can act on.

Two clustering approaches were applied and compared on the same dataset: K-Means (the production-scale approach) and Agglomerative Hierarchical Clustering (used here as a validation method on a sample). The end goal was a set of customer segments that are both statistically defensible and directly usable by marketing and risk teams.

Business Context

Customer segmentation is one of the most widely used machine learning applications in banking it allows a single customer base to be split into groups that receive different treatment: retention offers for dormant customers, loyalty perks for high-value spenders, and risk review for revolvers.

2. Dataset

Source: Credit Card Dataset for Clustering (Kaggle)

Size: 8,950 active credit card holders, 18 original features covering 6 months of behaviour

Feature themes: balance, purchase behaviour (one-off vs. installment), cash advance usage, credit limit, payment history, and account tenure

The CUST_ID column was dropped before modeling, as it is a unique string identifier that carries no behavioural signal and would only distort distance-based clustering.

3. Methodology

3.1 Data Preprocessing

- **Missing values:** 313 missing values in MINIMUM_PAYMENTS and 1 in CREDIT_LIMIT (314 of 152,150 cells) were filled with the column median rather than the mean, since both fields are right-skewed by a small number of very high-limit customers.
- **Feature scaling:** All 17 remaining features were standardized with StandardScaler (mean 0, standard deviation 1). This is mandatory for clustering because K-Means and hierarchical clustering both rely on Euclidean distance without scaling, a high-magnitude feature like CASH_ADVANCE would dominate the distance calculation over a 0–1 feature like BALANCE_FREQUENCY purely due to its numeric range, not its actual importance.

3.2 Choosing the Number of Clusters

K-Means was fit for every k from 2 to 10, recording inertia (within-cluster sum of squares) at each step, and the Silhouette Score which measures how well-separated clusters are — was computed alongside it for cross-validation of the choice.

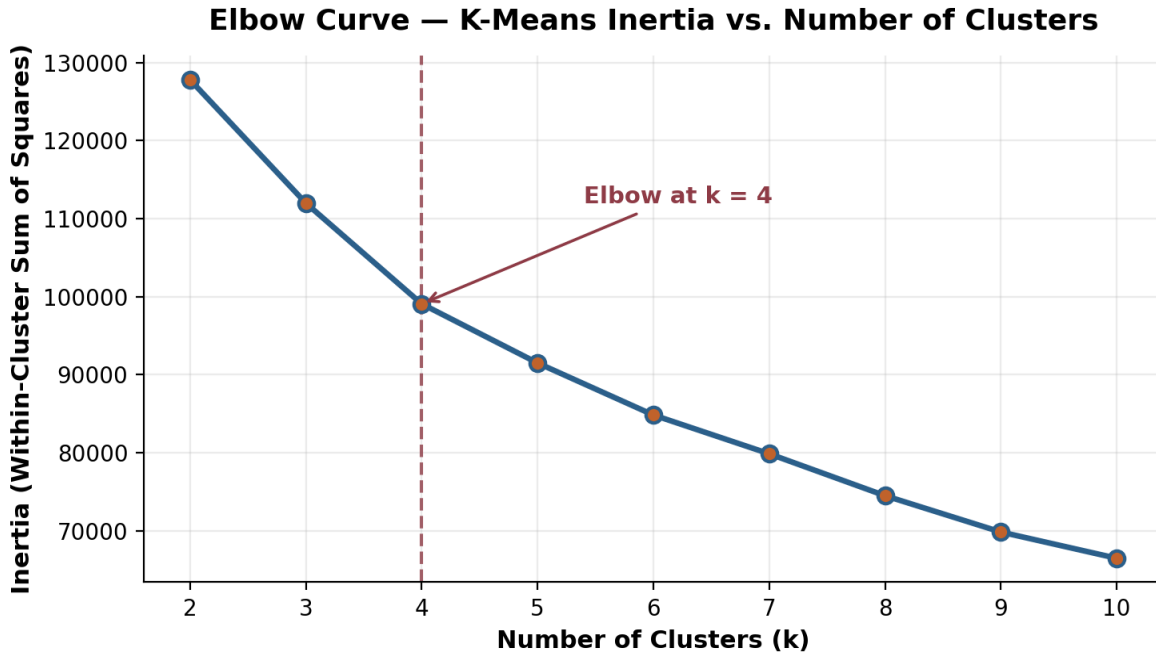


Figure 1 — Elbow Curve. Inertia drops sharply through $k = 4$, then the rate of improvement flattens noticeably.

The elbow curve bends at $k = 4-5$, while the silhouette score actually peaks at $k = 3$ (0.251) before dipping. This is a common, realistic disagreement: the elbow method measures overall compactness, while silhouette measures separation between clusters they don't always agree on the same k .

Decision: $k = 4$

$k = 4$ was selected as the final cluster count. It sits at the elbow bend, its silhouette score (0.198) remains reasonable and close to the $k = 3$ peak, and critically for the business use case — 4 segments give marketing and risk teams enough resolution to act on; $k = 3$ was found to merge two behaviourally distinct groups together.

4. Quantitative Results

4.1 Inertia and Silhouette Score by k

k	2	3	4	5	6	7	8	9	10
Inertia	127,784.5	111,975.0	99,061.9	91,490.5	84,826.6	79,856.2	74,484.9	69,828.7	66,466.4
Silhouette	0.210	0.251	0.198	0.193	0.203	0.208	0.222	0.226	0.220

4.2 Cluster Sizes (k = 4)

Customer Distribution Across Segments (n = 8,950)

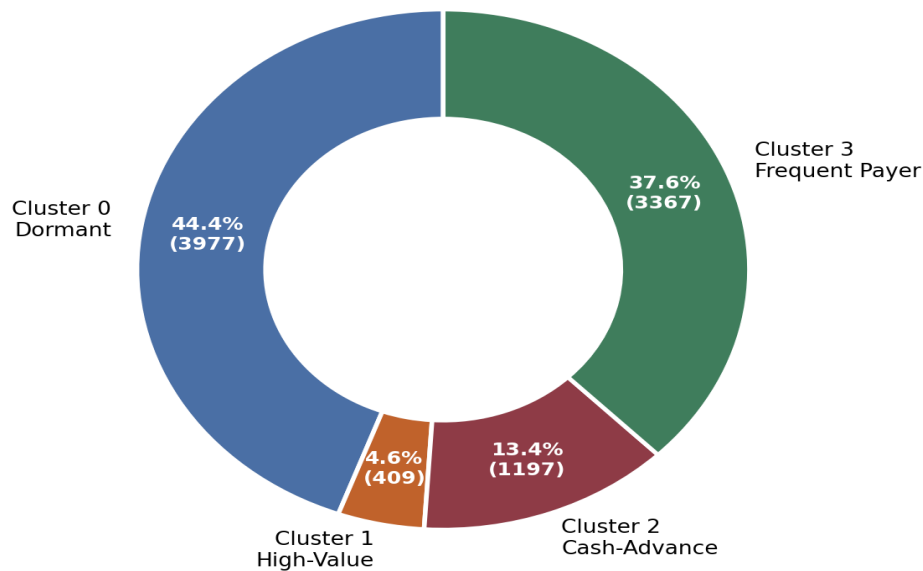


Figure 2 — Customer distribution across the four final segments (n = 8,950).

4.3 Cluster Profile — Mean Feature Values

Cluster	Balance	Purchases	Cash Advance	Credit Limit	Full Payment %
0 — Dormant (3,977)	1,012.7	270.0	596.5	3,278.6	0.10
1 — High-Value (409)	3,551.2	7,681.6	653.6	9,696.9	0.30
2 — Cash-Advance (1,197)	4,602.4	501.9	4,521.5	7,546.2	0.00
3 — Frequent Payer (3,367)	894.9	1,236.2	210.6	4,213.2	0.30

5. Cluster Interpretation

- Cluster 0 — Low-Activity / Dormant (3,977 customers, 44.4%):** Low balance, low purchases of any kind, low cash advance usage, and the lowest credit limits. These customers barely use their card. Recommended action: re-engagement campaigns and activation offers.
- Cluster 1 — High-Value Spenders (409 customers, 4.6%):** By far the highest purchases (especially one-off), highest credit limits and payments, high purchase frequency. Premium customers who spend heavily and pay it down. Recommended action: loyalty rewards, premium card upgrades, higher-limit cross-sell.

- **Cluster 2 — Cash-Advance Users / Risk Segment (1,197 customers, 13.4%):** High balance but low regular purchases, by far the highest cash advance amount and frequency, and the lowest full-payment rate (0%). This segment relies on cash withdrawals and carries revolving debt. Recommended action: credit risk review, interest-rate reassessment, financial wellness outreach.
- **Cluster 3 — Frequent Moderate Spenders / Good Payers (3,367 customers, 37.6%):** Low balance, moderate installment-driven purchases, high purchase frequency, and a healthy 30% full-payment rate. Everyday users who pay responsibly. Recommended action: loyalty programs and modest credit limit increases low risk, consistent engagement.

Taken together, the segmentation cleanly separates customers along two business-relevant dimensions: how they use credit (purchasing-driven vs. cash-advance-driven) and how well they manage it (full payers vs. revolvers) exactly the two axes a bank needs for targeted marketing (Clusters 0, 1, 3) versus risk management (Cluster 2).

6. Hierarchical Clustering — Validation

Because Agglomerative Hierarchical Clustering has $O(n^2)$ memory and time complexity, it was applied to a random sample of 300 customers from the scaled data using Ward's linkage the same variance-minimizing objective K-Means optimizes, making the two methods directly comparable. Cutting the resulting dendrogram at a distance of 28.82 produces exactly 4 clusters, confirming that the $k = 4$ structure found by K-Means also holds up hierarchically.

6.1 K-Means vs. Agglomerative — Cross-Tabulation (n = 300 sample)

K-Means \ Agglomerative	Agg. 0	Agg. 1	Agg. 2	Agg. 3
K-Means 0	1	6	124	0
K-Means 1	15	0	0	0
K-Means 2	1	31	6	1
K-Means 3	81	0	33	1

Most K-Means clusters have one dominant matching Agglomerative cluster for example, 124 of the customers K-Means placed in Cluster 0 also land in Agglomerative Cluster 2, and 81 of K-Means Cluster 3 land in Agglomerative Cluster 0 showing real structural agreement on the core of each segment. There is meaningful cross-over at the boundaries, which is expected: the two algorithms optimize different objectives (centroid distance vs. variance-minimizing merges), so customers near a boundary between behavioural groups get assigned differently depending on the method.

7. Recommendation

K-Means is recommended for production use.

It scales to the full ~9,000-customer base (hierarchical clustering's $O(n^2)$ cost is impractical beyond a few thousand rows), it is fast to re-run as new customers are added monthly, and its output one cluster label plus interpretable cluster-mean profiles plugs directly into marketing and risk workflows.

Hierarchical clustering remains valuable as a validation step, as demonstrated here, rather than as the production method.

8. Conclusion

This task moved from raw, unlabeled customer data to four statistically validated, business-interpretable segments Dormant, High-Value, Cash-Advance Risk, and Frequent Payer using K-Means as the primary method and hierarchical clustering as an independent structural check. The elbow method, silhouette analysis, and dendrogram all converged on the same $k = 4$ structure, and the resulting segments map cleanly onto concrete marketing and credit-risk actions a bank could take today.